

(a) Prove that

$$\sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta + \cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 4 \cos \theta \cos 2\theta (\sin 4\theta + \cos 4\theta).$$

(b) Solve the equation $\sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta + \cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$,

$$\text{where } 0 \leq \theta \leq \frac{\pi}{2}.$$

(6 marks)

$$\begin{aligned} \text{a)} \quad & \sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta + \cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta \\ &= 2 \sin 2\theta \cos \theta + 2 \sin 6\theta \cos \theta + 2 \cos 2\theta \cos \theta + 2 \cos 6\theta \cos \theta \quad \text{1M} \\ &= 2 \cos \theta (\sin 2\theta + \sin 6\theta + \cos 2\theta + \cos 6\theta) \\ &= 2 \cos \theta (2 \sin 4\theta \cos 2\theta + 2 \cos 4\theta \cos 2\theta) \quad \text{1M} \\ &= 4 \cos \theta \cos 2\theta (\sin 4\theta + \cos 4\theta) \quad \text{1A} \end{aligned}$$

$$\text{b)} \quad \sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta + \cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$$

$$4 \cos \theta \cos 2\theta (\sin 4\theta + \cos 4\theta) = 0 \quad \text{1M}$$

$$\cos \theta \text{ or } \cos 2\theta \text{ or } \sin 4\theta + \cos 4\theta = 0$$

$$\cos \theta = 0 \text{ or } \cos 2\theta = 0 \text{ or } \tan 4\theta = -1$$

$$\theta = \frac{\pi}{2} \text{ or } \theta = \frac{\pi}{4} \text{ or } 4\theta = \frac{3\pi}{4}, \frac{7\pi}{4} \quad \text{1M}$$

$$\theta = \frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{16}, \frac{7\pi}{16} \quad \text{1A for all}$$